EXPERIMENT 10

INPUT SIGNAL

```
clc;
clear;
close all;

% Given value
a = 0.4;

% Number of samples
n = 0:20;

% Input: Unit step signal
x = ones(1,length(n));

% Difference equation
y = zeros(1,length(n));

for k = 2:length(n)
    y(k) = a*y(k-1) + x(k);
end

% Steady-state value
yss = 1/(1-a);

fprintf('Steady State Value = %.4f\n', yss);

% Plot output response
stem(n,y,'filled');
hold on;
yline(yss,'r--','Steady State');
title('Step Response of Stable DT System');
xlabel('Sample Number (n)');
ylabel('Amplitude');
legend('Output Response','Steady State');
grid on;
```

OUTPUT

```
Steady State Value = 1.6667
```

